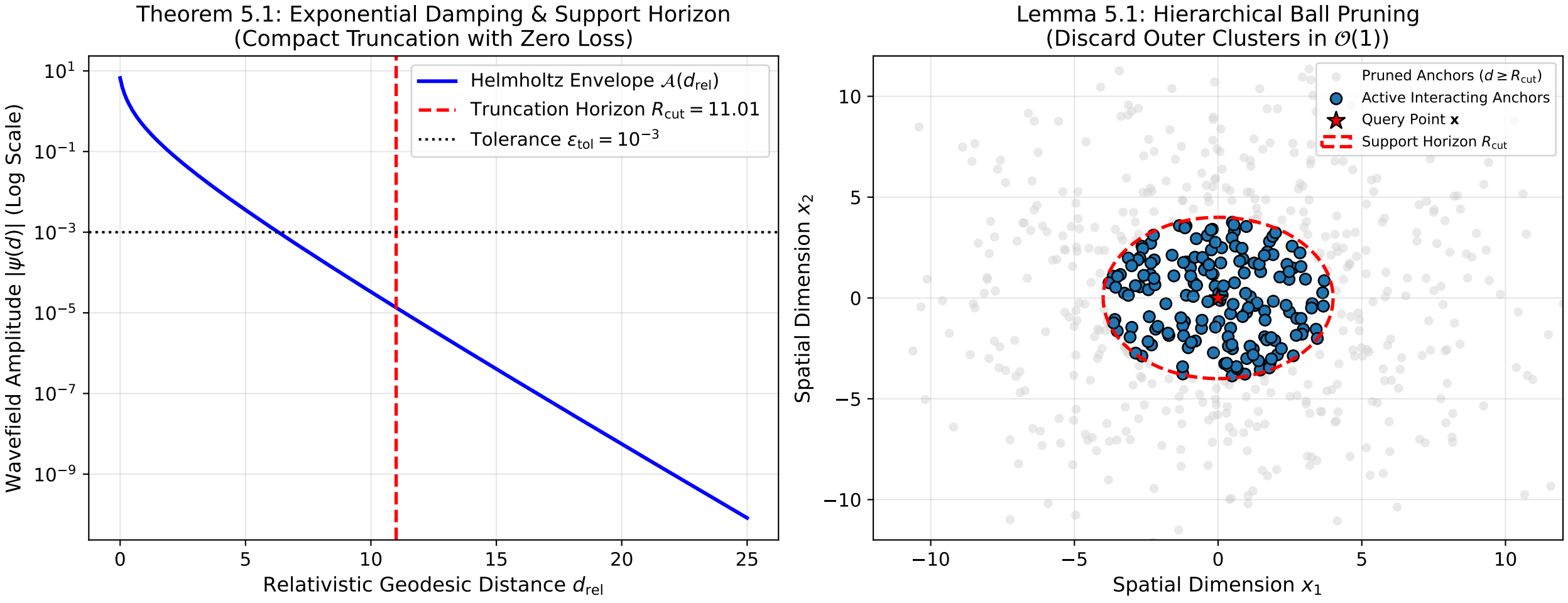
Theorem 5.1: Exponential Damping & Support Horizon
(Compact Truncation with Zero Loss)
Helmholtz Envelope $\mathcal{A}(d_{\text{rel}})$
Truncation Horizon $R_{\text{cut}} = 11.01$
Tolerance $\varepsilon_{\text{tol}} = 10^{-3}$
Wavefield Amplitude $|\psi(d)|$ (Log Scale)
Relativistic Geodesic Distance d_{rel}
Lemma 5.1: Hierarchical Ball Pruning
(Discard Outer Clusters in $\mathcal{O}(1)$)
Pruned Anchors ($d \geq R_{\text{cut}}$)
Active Interacting Anchors
Query Point $\mathbf{x}$
Support Horizon R_{cut}
Spatial Dimension x_1
Spatial Dimension x_2